

Sharp Subexponential Reversion

Exact extremal cost, a Gevrey transition,
and composition of two divergent series

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Abstract

Exact preservation of a weighted coefficient type under formal reversion is already established in the inspected ProveIt corpus. It does not quantify the subexponential amplification at the critical coefficient radius. We address that subsequent question by solving a coefficientwise extremal problem for two factorial normalizations. A single negative series attains the worst inverse coefficient simultaneously at every degree. For its normalized inverse amplification R_n , we prove $\log R_n \sim Cn^{1-s}$ for every $0 < s < 1$, a multiplicative equivalent $R_n \sim \exp(Cn^{1-s})$ for $s > 1/2$, and the exact additional constant at $s = 1/2$. A uniform equivalent yields the crossover $R_n \rightarrow \exp(Ce^{-t})$ when $s = 1+t/\log n$. Consequently, fixed-radius factorial coefficient balls are stable under inversion exactly for $s \geq 1$, despite exact type preservation at every positive order.

A second result answers the Gevrey version of the repository's question about two divergent composition arguments: their types obey a sharp maximum inequality, with equality whenever the two competing types are unequal. Finally, an explicit positive-coefficient series has a full normalized root limit although its inverse has infinitely many zero coefficients. All proofs are supplied. Finite exact checks and numerical diagnostics are distinguished from proofs and from Lean verification.

Scope and status. Conventional mathematical proofs and reproducible finite calculations. No Lean formalization, analytic summation theorem, or worldwide priority claim is asserted. The contribution is measured against explicitly identified questions in the inspected repository.

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1 Research target and contribution boundaries

1.1 Which repository questions are actually still being extended?

The source for this continuation is the ProveIt package *Sharp Weighted Type Under Formal Reversion: zero-loss inversion, exact Borel radii, and non-D-finiteness in countable exponential feedback* [1, 3]. In its section on further research, the subsection “Optimal subexponential cost of reversion” asks for the smallest universal overhead at fixed weight and fixed input constants, naming $M_n = (n!)^s$, $0 < s < 1$, as a first target. The subsection “Two divergent composition arguments” asks for a filtered composition estimate and a noncancellation condition when both arguments are divergent. These are the two concrete questions treated here.

The inspection used the main-branch snapshot identified by

5c695cfd f70a8b6c 91bd5b2c 1e99e3ba ecd5a864

The relevant source file is

Analysis/Transseries/docs/series-and-transseries/
Sharp_Weighted_Type_Formal_Reversion/article.tex.

Its research questions occur in the inspected range 1418–1515; the individual subsection headings, rather than mutable line numbers, are the durable locators. The directory inventory and editorial notes were also consulted [2].

There is a significant duplication trap. The amplitude–slope package [4] asks whether inverse and forward Gevrey types agree. The current inventory explicitly observes that the weighted-type theorem already answers that question. Accordingly, merely reproving equality of these types would not be a new repository contribution. The present work moves below the n th-root scale and treats two genuinely divergent composition inputs. The older type-invariance statement is reproved only as a short, self-contained tool.

1.2 Results and exact extent of the resolution

The principal result, [theorem 2.2](#), identifies the exact finite-degree extremum and its leading subexponential asymptotics. For $0 < s < 1$, the coefficient in the exponent is determined, not just the power of n . For $s \geq 1/2$, the statements give a full multiplicative equivalent, including the separate constant at the boundary $s = 1/2$. For $0 < s < 1/2$, a complete multiplicative equivalent remains open here: further subexponential terms cannot in general be discarded.

The maximum law in [theorem 8.1](#) settles the factorial-weight version of the second question. It does not settle the same problem for every superexponential log-convex weight. Equal-type cancellation is unavoidable; unequal-type inputs supply an explicit sufficient noncancellation condition.

The positive-forward counterexample in [theorem 9.1](#) explains why exact type alone cannot be strengthened to a full inverse root limit, even after assuming positivity and a full root limit on the forward side. It is a general obstruction, not a claim about the signs of a particular exponential-feedback inverse.

1.3 Relation to established literature

Formal Lagrange inversion is classical; Gessel’s survey supplies a systematic account of its coefficient forms and proofs [5]. Composition and inverse stability in ultradifferentiable classes

have been studied by Rainer and Schindl [7, 8]. Borinsky develops composition and inversion in a factorially divergent coefficient-asymptotic algebra [6]. These precedents are relevant: neither Gevrey closure nor the usefulness of a single large coefficient should be advertised as newly discovered here.

Our object is a sharp extremum over a *fixed formal coefficient ball*, including its dependence on the order s , rather than stability of a function class after increasing an unspecified exponential constant. The proof is elementary and includes the exact constants needed for this object. The literature check was targeted, not an exhaustive priority search; the results should be independently reviewed before publication or incorporation as established repository theorems.

Assertion	Status in this article
Coefficientwise extremal inverse	Exact finite identity and proof.
Optimal logarithmic cost, $0 < s < 1$	Upper and lower bounds with the same constant.
Multiplicative cost, $s > 1/2$	Proved, locally uniformly in s and C .
Boundary $s = 1/2$	Proved with its additional constant.
Two divergent composition inputs	Sharp type inequality; equality at unequal types.
General type invariance	Existing repository result; factorial case reproved.
Full feedback inverse asymptotics	Not inferred from a universal upper envelope.
Computations	Exact finite tests and non-certified floating-point diagnostics.

Table 1: Dependency and claim ledger. None of the mathematical results in this article is presented as Lean-verified.

2 The extremal problem and the main theorem

2.1 Two normalizations, kept explicit

Fix $s > 0$, $A > 0$, $C > 0$, and $\nu \in \{0, 1\}$. For $j \geq 1$, define

$$w_j = w_j(s, \nu) = \left(\frac{(j + \nu)!}{(1 + \nu)!} \right)^s. \quad (2.1)$$

In particular,

$$w_1 = 1, \quad w_2 = (\nu + 2)^s, \quad \frac{w_{j+1}}{w_j} = (j + \nu + 1)^s. \quad (2.2)$$

The symbol w_0 is not needed. When it is useful in a computational recurrence, we set $w_0 = 1$ as a separate convention.

Definition 2.1 (A critical-radius coefficient ball). Let $\mathcal{B}_{s,A,C}^{(\nu)}$ be the set of all formal series

$$f(z) = z + \sum_{j \geq 1} a_{j+1} z^{j+1} \in \mathbb{C}[[z]], \quad |a_{j+1}| \leq CA^j w_j. \quad (2.3)$$

For $g = f^{(-1)}$, define the normalized extremal amplification

$$R_n(s, C, \nu) = \sup_{f \in \mathcal{B}_{s,A,C}^{(\nu)}} \frac{|[z^n] f^{(-1)}|}{CA^{n-1} w_{n-1}}, \quad n \geq 2. \quad (2.4)$$

The theorem below shows that this expression is independent of A . The inverse is compositional, not the reciprocal $1/f$.

The case $\nu = 0$ uses $(n-1)!^s$ for the n th coefficient. The case $\nu = 1$ uses $(n!/2)^s$. Thus the conventional ball

$$|a_n| \leq BA^{n-1}(n!)^s \quad (n \geq 2) \quad (2.5)$$

is exactly (2.3) with $\nu = 1$ and $C = 2^s B$. The two weights have identical n th-root types but different fixed-ball constants. This difference matters in every sharp subexponential formula.

Theorem 2.2 (Sharp extremal reversion cost). *Let s, A, C, ν be as above. Define*

$$F_*(z) = z - C \sum_{j \geq 1} A^j w_j z^{j+1}, \quad G_* = F_*^{(-1)}. \quad (2.6)$$

All coefficients of G_ after the constant term are strictly positive. The same series F_* attains the supremum in (2.4) for every n . Moreover:*

(i) *For every fixed $0 < s < 1$,*

$$\lim_{n \rightarrow \infty} \frac{\log R_n(s, C, \nu)}{n^{1-s}} = C. \quad (2.7)$$

(ii) *For every fixed $s > 1/2$,*

$$R_n(s, C, \nu) = \exp(Cn^{1-s}) (1 + o(1)). \quad (2.8)$$

The $o(1)$ is uniform for s in a compact subset of $(1/2, \infty)$ and C in a compact subset of $(0, \infty)$.

(iii) *At $s = 1/2$,*

$$R_n(1/2, C, \nu) = \exp\left(C\sqrt{n} + \frac{3C^2}{4} + C\sqrt{\nu+2}\right) (1 + o(1)). \quad (2.9)$$

(iv) *Uniformly for t in every bounded real interval,*

$$R_n\left(1 + \frac{t}{\log n}, C, \nu\right) \rightarrow \exp(Ce^{-t}). \quad (2.10)$$

The theorem gives the following sharp classification:

order	extremal amplification
$s > 1$	$R_n \rightarrow 1$
$s = 1$	$R_n \rightarrow e^C$
$0 < s < 1$	$R_n \rightarrow \infty, \quad \log R_n \sim Cn^{1-s}.$

(2.11)

At $0 < s < 1$, this is an amplification at the *same* A ; it is still subexponential and hence invisible to the coefficient type.

Corollary 2.3 (The unshifted factorial convention). *For the ball (2.5), normalize inverse coefficients by $BA^{n-1}(n!)^s$ and call their extremum $\tilde{R}_n$. Then, for $0 < s < 1$,*

$$\log \tilde{R}_n \sim 2^s B n^{1-s}.$$

For $s > 1/2$, $\tilde{R}_n \sim \exp(2^s B n^{1-s})$. At $s = 1/2$,

$$\tilde{R}_n \sim \exp\left(\sqrt{2}B\sqrt{n} + \frac{3}{2}B^2 + \sqrt{6}B\right).$$

In particular, $\tilde{R}_n \rightarrow e^{2B}$ when $s = 1$.

Proof. Use $\nu = 1$ and $C = 2^s B$ in theorem 2.2. □

3 Finite coefficient algebra and uniform convolution bounds

3.1 An exact simultaneous extremizer

For $f = z(1 + v)$ with $v \in z\mathbb{C}[[z]]$, formal Lagrange inversion gives

$$[z^n]f^{\langle -1 \rangle} = \frac{1}{n}[z^{n-1}](1 + v(z))^{-n}. \quad (3.1)$$

Every coefficient is a finite sum; no radius of convergence is needed. Since

$$(1 + v)^{-n} = \sum_{k \geq 0} (-1)^k \binom{n + k - 1}{k} v^k,$$

taking absolute values and then replacing each $|[z^j]v|$ by its upper bound in (2.3) gives equality when all coefficients of v are negative and saturate the bounds.

For $m \geq k \geq 1$, put

$$P(m, k) = \sum_{\substack{j_1 + \dots + j_k = m \\ j_i \geq 1}} \prod_{i=1}^k w_{j_i}. \quad (3.2)$$

Also set $P(0, 0) = 1$ and $P(m, 0) = 0$ for $m > 0$. Then the exact value of the extremum is

$$R_n = \frac{1}{nw_{n-1}} \sum_{k=1}^{n-1} \binom{n + k - 1}{k} C^{k-1} P(n - 1, k). \quad (3.3)$$

All powers of A cancel because the indices in each composition sum to $n - 1$. The $k = 1$ summand is exactly 1, so $R_n \geq 1$. This proves all finite extremizer assertions of [theorem 2.2](#).

3.2 A factorial convolution estimate

Lemma 3.1 (Uniform two-part convolution). *There is a finite $K = K(s, \nu) \geq 1$ such that*

$$\sum_{j=1}^{m-1} w_j w_{m-j} \leq K w_{m-1} \quad (m \geq 2). \quad (3.4)$$

The constant can be chosen bounded when s varies in any compact subset of $(0, \infty)$. Consequently,

$$P(m, k) \leq K^{k-1} w_{m-k+1} \quad (m \geq k \geq 1). \quad (3.5)$$

Proof. For $1 \leq j \leq m/2$,

$$\binom{m}{j} = \frac{m}{j} \binom{m-1}{j-1} \geq \frac{m}{j} 2^{j-1}.$$

Indeed, each factor in the product for $\binom{m-1}{j-1}$ is at least 2 when $m \geq 2j$. For $\nu = 0$ this gives

$$\frac{w_j w_{m-j}}{w_{m-1}} = \left(\frac{m}{\binom{m}{j}} \right)^s \leq \left(\frac{j}{2^{j-1}} \right)^s.$$

For $\nu = 1$ the ratio is

$$\left(\frac{(j+1)(m-j+1)}{2 \binom{m}{j}} \right)^s \leq \left(\frac{j(j+1)}{2^j} \right)^s.$$

Reflect about $m/2$ and sum. Twice the corresponding infinite sum is a valid K after increasing it to at least 1. Both sums converge uniformly on compact positive s -intervals.

To prove (3.5), use induction on k . Splitting off the first part gives

$$P(m, k) \leq K^{k-2} \sum_{j=1}^{m-k+1} w_j w_{m-k+2-j} \leq K^{k-1} w_{m-k+1}.$$

The case $k = 1$ is an equality. □

Let $T_{n,k}$ denote the k th summand in (3.3). For $k \leq n/2$, theorem 3.1 implies

$$T_{n,k} \leq 2 \frac{(Dn^{1-s})^{k-1}}{k!}, \quad D = 2^{s+1}CK. \quad (3.6)$$

In fact, $\binom{n+k-1}{k} \leq (2n)^k/k!$, and the ratio w_{n-k}/w_{n-1} contains $k-1$ factors bounded by $(n/2)^{-s}$. For $k > n/2$, at least $\lfloor n/2 \rfloor - 1$ such factors remain, whereas the binomial coefficient is at most 4^n . Thus

$$\sum_{k > n/2} T_{n,k} \leq \exp\left(-\frac{s}{2}n \log n + O(n)\right). \quad (3.7)$$

Here and below constants are allowed to depend on fixed s, C, ν . They are uniform on compact positive s, C sets when that is asserted. Together these estimates give

$$1 \leq R_n \leq 2 \exp(Dn^{1-s}) + \exp\left(-\frac{s}{2}n \log n + O(n)\right). \quad (3.8)$$

This already proves subexponential overhead. Determining its sharp constant requires more than replacing every convolution by K .

4 Localization at one large coefficient

A positive composition of $m = n - 1$ is called *localized at scale h* if one of its parts is at least $m - h$, where $h < m/2$. Such a part is unique. This terminology concerns the finite Lagrange sum only.

Lemma 4.1 (Negligible configurations without a large part). *Let the contribution to (3.3) of nonlocalized compositions be $E_n(h)$. Then:*

(i) *For fixed $0 < \epsilon < 1/2$ and $h = \lfloor \epsilon m \rfloor$, $E_n(h) \leq \exp(-c_\epsilon n)$ for all sufficiently large n .*

(ii) *If*

$$\max(1 - s, 0) < \kappa < \alpha < 1, \quad h = \lfloor n^\alpha \rfloor, \quad (4.1)$$

then $E_n(h) = o(1)$. More precisely it is bounded by the sum of $\exp(-cn^\alpha \log n)$ and $\exp(-c'n^\kappa \log n)$.

Proof. First restrict to compositions of length at most $t = o(n)$. If every part is at most $m - h$ and $m - h \geq m/2$, elementary factorial concentration gives

$$\prod_i j_i! \leq (m - h)!h!. \quad (4.2)$$

To see this, allow zero parts and transfer units from a smaller positive part to a larger one, without exceeding $m - h$. Each transfer weakly increases the factorial product. A maximizer has at most two positive parts, of sizes $m - h$ and h .

For $\nu = 0$, the normalized weight product is therefore at most $\binom{m}{h}^{-s}$. For $\nu = 1$, there is an additional factor bounded by

$$\left(\prod_i (j_i + 1) \right)^s \leq (1 + m/k)^{sk}$$

for a composition of length k , by the arithmetic–geometric mean inequality. The factors 2^{-sk} and the extra normalization in w_m only improve this upper bound.

There are $\binom{m-1}{k-1}$ compositions of length k . Both this count and $\binom{n+k-1}{k}$ have logarithm $O(k \log(en/k))$. The powers of C have logarithm $O(k)$. Summing over $k \leq t$ consequently bounds this portion by

$$\exp \left\{ -s \log \binom{m}{h} + O \left(t \log \frac{en}{t} + \log n \right) \right\}. \quad (4.3)$$

The function $k \log(en/k)$ is increasing for the relevant k .

Lengths $k > t$ are controlled by (3.6) and (3.7). If $t/n^{1-s} \rightarrow \infty$ and $t \leq n/2$, the small-length tail is bounded, after harmless polynomial factors, by

$$\exp \left\{ -t \log \frac{t}{eDn^{1-s}} \right\}. \quad (4.4)$$

This follows either from successive-term ratios or from $k! \geq (k/e)^k$.

For (i), choose $t = \lfloor n/\sqrt{\log n} \rfloor$. The error term in (4.3) is $o(n)$, while $\log \binom{m}{h} = nH(\epsilon) + o(n)$, where $H(x) = -x \log x - (1-x) \log(1-x) > 0$. The exponent in (4.4) is at most $-(s + o(1))n\sqrt{\log n}$. For (ii), take $t = \lfloor n^\kappa \rfloor$. Then $\log \binom{m}{h} = (1 - \alpha + o(1))n^\alpha \log n$ dominates $O(n^\kappa \log n)$, and (4.4) has a negative constant multiple of $n^\kappa \log n$. This proves both statements. $\square$

4.1 The exact localized sum

Write $m = n - 1$ and let $R_n^{\text{loc}}(h)$ be the localized contribution. Let the large part have size $m - d$, let ℓ be the number of other parts, and let m_j count the other parts of size j . Then

$$\ell = \sum_{j \geq 1} m_j, \quad d = \sum_{j \geq 1} j m_j. \quad (4.5)$$

Choosing the position of the large part changes the Lagrange factor by

$$\frac{\ell + 1}{n} \binom{n + \ell}{\ell + 1} = \binom{n + \ell}{\ell}.$$

Thus, exactly,

$$R_n^{\text{loc}}(h) = \sum_{\substack{m_1, m_2, \dots \geq 0 \\ d \leq h}} \binom{n + \ell}{\ell} \frac{\ell!}{\prod_j m_j!} C^\ell \frac{w_{m-d}}{w_m} \prod_j w_j^{m_j}. \quad (4.6)$$

Only $j \leq h$ can occur. The all-zero multiplicity vector contributes 1. Formula (4.6) is the source of all subsequent constants.

5 Optimal logarithmic amplification for every $0 < s < 1$

5.1 Lower bound from quadratic decorations

In (4.6), retain only parts of size 1 beside the large part. Their contribution for a given ℓ is

$$L_{n,\ell} = \binom{n+\ell}{\ell} C^\ell \frac{w_{n-1-\ell}}{w_{n-1}}. \quad (5.1)$$

Take $\ell = \lfloor Cn^{1-s} \rfloor$. It tends to infinity and is $o(n)$. Uniform elementary product estimates give

$$\log \binom{n+\ell}{\ell} = \ell \log n - \log(\ell!) + O(\ell^2/n + \ell/n), \quad (5.2)$$

$$\log \frac{w_{n-1-\ell}}{w_{n-1}} = -s\ell \log n + O(\ell^2/n + \ell/n). \quad (5.3)$$

Stirling's estimate then yields

$$\log L_{n,\ell} = \ell \log \frac{Cn^{1-s}}{\ell} + \ell + o(\ell) = Cn^{1-s} + o(n^{1-s}).$$

For large n the part $n-1-\ell$ is genuinely the unique large part, so these are actual distinct summands, not a formal overcount. Hence

$$\liminf_{n \rightarrow \infty} \frac{\log R_n}{n^{1-s}} \geq C. \quad (5.4)$$

5.2 Upper bound with the same constant

Fix $0 < \epsilon < 1/2$, let $h = \lfloor \epsilon m \rfloor$, and put

$$t_n = (m + \nu - h + 1)^{-s}. \quad (5.5)$$

For $d \leq h$, each of the d factors in the factorial quotient gives

$$\frac{w_{m-d}}{w_m} \leq t_n^d.$$

Define the finite polynomial $H_h(t) = \sum_{j=1}^h w_j t^j$. Extending the positive sum over the small parts and then over ℓ in (4.6) gives

$$R_n^{\text{loc}}(h) \leq \sum_{\ell \geq 0} \binom{n+\ell}{\ell} (CH_h(t_n))^\ell = (1 - CH_h(t_n))^{-n-1}. \quad (5.6)$$

The last identity is used only when n is sufficiently large that $CH_h(t_n) < 1$.

For $1 \leq j < h$, successive terms of $H_h(t_n)$ have ratio $(j + \nu + 1)^s t_n$. Its maximum tends to $(\epsilon/(1-\epsilon))^s < 1$. Therefore

$$H_h(t_n) = t_n + O(t_n^2), \quad t_n = (1 - \epsilon)^{-s} n^{-s} (1 + o(1)).$$

Taking logarithms in (5.6) proves

$$\log R_n^{\text{loc}}(h) \leq C(1 - \epsilon)^{-s} n^{1-s} + o(n^{1-s}).$$

The nonlocalized contribution is exponentially small by [theorem 4.1\(i\)](#), whereas the localized contribution is at least 1. Consequently,

$$\limsup_{n \rightarrow \infty} \frac{\log R_n}{n^{1-s}} \leq C(1 - \epsilon)^{-s}.$$

Letting $\epsilon \downarrow 0$ and using (5.4) proves (2.7). In particular, no universal bound $\exp(o(n^{1-s}))$ is possible for this fixed ball, and no smaller leading constant than C can hold.

6 Multiplicative asymptotics and the boundary constant

6.1 A finite Poisson representation

The probability notation used here is just a convenient normalization of a finite positive sum. Fix a cutoff $h = o(n)$ and define

$$\lambda_j = Cw_j n^{1-sj} \quad (1 \leq j \leq h), \quad \Lambda_n = \sum_{j=1}^h \lambda_j. \quad (6.1)$$

Let N_j be independent Poisson variables of means λ_j and put

$$L = \sum_{j=1}^h N_j, \quad D_0 = \sum_{j=1}^h j N_j. \quad (6.2)$$

The subscript on D_0 distinguishes this random total size from the constant D in (3.6).

For a multiplicity vector of length ℓ and size d , extract n^ℓ from $(n+1) \cdots (n+\ell)$ and n^{-sd} from the large weight quotient in (4.6). If $d \leq h$,

$$\log \left\{ \frac{(n+1) \cdots (n+\ell)}{n^\ell} \frac{w_{n-1-d}}{w_{n-1}} n^{sd} \right\} = O \left(\frac{h^2 + h}{n} \right). \quad (6.3)$$

The constants are uniform on compact s -intervals. Thus, when $h^2/n \rightarrow 0$,

$$R_n^{\text{loc}}(h) = (1 + o(1)) e^{\Lambda_n} \mathbb{P}(D_0 \leq h). \quad (6.4)$$

No probabilistic limit theorem is hidden in this identity: it follows by multiplying and dividing each summand by $\prod_j e^{-\lambda_j}$.

6.2 Proof of the equivalent above one half

Assume first that $s > 1/2$. Choose

$$\max(1-s, 0) < \kappa < \alpha < 1/2, \quad h = \lfloor n^\alpha \rfloor.$$

Nonlocalized configurations contribute $o(1)$ by theorem 4.1(ii). For the intensities in (6.1), successive terms have ratio

$$\frac{\lambda_{j+1}}{\lambda_j} = \left(\frac{j + \nu + 1}{n} \right)^s.$$

Uniformly for $j < h$, this tends to zero. Since $w_1 = 1$,

$$\Lambda_n = Cn^{1-s} + O(n^{1-2s}), \quad \mathbb{E}D_0 = Cn^{1-s} + O(n^{1-2s}). \quad (6.5)$$

The error in the first formula tends to zero. The mean in the second is $o(h)$, so Markov's inequality gives $\mathbb{P}(D_0 \leq h) \rightarrow 1$. Formula (6.4) proves (2.8).

For a compact interval of s -values with lower endpoint $s_0 > 1/2$, choose κ, α once with $\max(1-s_0, 0) < \kappa < \alpha < 1/2$. Every estimate just used is then uniform, as are the convolution constants. Compact variation in $C > 0$ is harmless. This proves the asserted local uniformity.

At $s_n = 1 + t/\log n$, $n^{1-s_n} = e^{-t}$ exactly. For bounded t , s_n eventually belongs to a fixed compact subinterval of $(1/2, \infty)$. The uniform formula gives (2.10).

6.3 Why the order-one-half constant is different

At $s = 1/2$, the typical number of small parts is of order $\sqrt{n}$. Their squared total size divided by n is no longer negligible. In addition, parts of size 2 have a nonvanishing Poisson mean.

Choose

$$\frac{1}{2} < \kappa < \alpha < \frac{2}{3}, \quad h = \lfloor n^\alpha \rfloor. \quad (6.6)$$

Localization again leaves an absolute $o(1)$ remainder. Expansion of the two finite products in (6.3) now gives, uniformly for $\ell \leq d \leq h$,

$$\log \frac{(n+1) \cdots (n+\ell)}{n^\ell} = \frac{\ell(\ell+1)}{2n} + O(h^3/n^2), \quad (6.7)$$

$$\log \left(\frac{w_{n-1-d}}{w_{n-1}} n^{sd} \right) = \frac{s\{d^2 + (1-2\nu)d\}}{2n} + O(h^3/n^2). \quad (6.8)$$

The linear terms are $o(1)$, and $h^3/n^2 = o(1)$. Hence

$$R_n^{\text{loc}}(h) = e^{\Lambda_n} \mathbb{E} \left[\exp \left(\frac{L^2 + sD_0^2}{2n} + o(1) \right) \mathbf{1}_{\{D_0 \leq h\}} \right], \quad (6.9)$$

where the $o(1)$ inside the expectation is uniform on the indicated event.

Here

$$\lambda_1 = C\sqrt{n}, \quad \lambda_2 = C\sqrt{\nu+2}, \quad \sum_{j=3}^h j^r \lambda_j = O(n^{-1/2}) \quad (r = 0, 1, 2). \quad (6.10)$$

It follows that

$$\Lambda_n = C\sqrt{n} + C\sqrt{\nu+2} + o(1), \quad \frac{L}{\sqrt{n}}, \frac{D_0}{\sqrt{n}} \longrightarrow C \quad \text{in probability.} \quad (6.11)$$

For example, these convergences follow directly from the corresponding means and variances, which are $C\sqrt{n} + O(1)$.

We must justify passage to the expectation of an exponential; convergence in probability alone would not suffice. Fix $\theta > 0$. Successive intensities after multiplication by $e^{\theta j}$ still have ratio $o(1)$, so

$$\mathbb{E} e^{\theta D_0} = \exp \left(C(e^\theta - 1)\sqrt{n} + O(1) \right). \quad (6.12)$$

On $D_0 \leq h$, we have $L \leq D_0$ and

$$\frac{L^2 + sD_0^2}{2n} \leq \frac{(1+s)hD_0}{2n} \leq \frac{\theta}{2} D_0$$

for all sufficiently large n . Consequently, for any sufficiently large fixed K ,

$$\begin{aligned} & \mathbb{E} \left[\exp \left(\frac{L^2 + sD_0^2}{2n} \right) \mathbf{1}_{\{K\sqrt{n} < D_0 \leq h\}} \right] \\ & \leq e^{-\theta K\sqrt{n}/2} \mathbb{E} e^{\theta D_0} = \exp \left(\{C(e^\theta - 1) - \theta K/2\}\sqrt{n} + O(1) \right) \longrightarrow 0. \end{aligned}$$

On $D_0 \leq K\sqrt{n}$, the integrand is bounded and converges in probability to $\exp((1+s)C^2/2)$. Thus the expectation in (6.9) tends to $\exp(3C^2/4)$. Combining this with (6.11) proves (2.9).

Remark 6.1 (The two sources of the boundary constant). The term $3C^2/4$ is the combined curvature correction from the rising binomial product and the depleted factorial of the large part. The term $C\sqrt{\nu+2}$ is the contribution of size-two small parts. Omitting either mechanism gives a wrong multiplicative equivalent. The normalization shift ν changes the second contribution, but not the first.

7 Fixed-radius stability and exact coefficient type

7.1 A sharp fixed-ball threshold

Corollary 7.1 (Fixed-radius boundedness). *For fixed s, A, C, ν , there is a finite C' such that*

$$f \in \mathcal{B}_{s,A,C}^{(\nu)} \implies |[z^n]f^{(-1)}| \leq C' A^{n-1} w_{n-1} \quad (n \geq 2)$$

if and only if $s \geq 1$.

Proof. The assertion is equivalent to boundedness of R_n . For $s \geq 1$, (2.11) supplies a finite limit, and each of the finitely many initial R_n is finite. For $0 < s < 1$, the single extremizer F_* has $R_n \rightarrow \infty$. $\square$

The threshold does not mean that Gevrey inversion fails below order one. For every $s > 0$ and every $A' > A$, (3.8) implies a bound at radius parameter A' with a constant independent of n . What fails below one is retaining the *same* A with only a constant prefactor.

7.2 Type preservation, included as a tool

For $f(z) = \sum f_n z^n$, define

$$\tau_s(f) = \limsup_{n \rightarrow \infty} \left(\frac{|f_n|}{(n!)^s} \right)^{1/n} \in [0, \infty]. \quad (7.1)$$

The convention $0^{1/n} = 0$ is understood. Finite initial coefficients do not affect this value. Replacing $(n!)^s$ by $((n-1)!)^s$ does not alter it either, since $n^{s/n} \rightarrow 1$.

Proposition 7.2 (Exact inverse type; a known repository result). *For $s > 0$ and $f(z) = az + O(z^2)$ with $a \neq 0$,*

$$\tau_s(f^{(-1)}) = \frac{\tau_s(f)}{|a|}. \quad (7.2)$$

The equality includes zero and infinite types.

Proof. First suppose $a = 1$ and $\tau_s(f) < \infty$. For every $A > \tau_s(f)$, there is a finite C with $|f_{j+1}| \leq CA^j (j!)^s$. This follows from the definition of limsup; the change of index and the polynomial factor $(j+1)^s$ can be absorbed in the strict exponential margin. The majorant (3.8) gives $\tau_s(f^{(-1)}) \leq A$. Let $A \downarrow \tau_s(f)$, including arbitrarily small positive A when the type is zero. Applying the same inequality to $f^{(-1)}$ proves the reverse inequality. If f has infinite type but its inverse had finite type, this finite-type argument applied to the inverse would be a contradiction.

For general a , let $F = f/a$. It is tangent to the identity and has the same type as f . Since $f^{(-1)}(w) = F^{(-1)}(w/a)$, input scaling gives (7.2). $\square$

The general log-convex theorem in [3] is broader than [theorem 7.2](#). The point of the present proof is that it exhibits the precise information that the type discards.

7.3 Borel radii, without a summability claim

For the coefficient normalization

$$\mathcal{B}_s f(\xi) = \sum_{n \geq 0} \frac{f_n}{(n!)^s} \xi^n,$$

Cauchy–Hadamard gives radius $1/\tau_s(f)$, with the usual conventions at zero and infinity. The frequently used normalization

$$\mathfrak{B}_s f(\xi) = \sum_{n \geq 0} \frac{f_n}{\Gamma(1 + sn)} \xi^n$$

has radius $s^s/\tau_s(f)$, because Stirling’s estimate gives

$$\left(\frac{(n!)^s}{\Gamma(1 + sn)} \right)^{1/n} \longrightarrow s^{-s}.$$

Neither radius determines analytic continuation, growth along a ray, or the existence of a Laplace integral. None of the extremal results above proves Borel summability or resurgence of the extremizer or of an exponential-feedback series.

8 Composition of two divergent series

Theorem 8.1 (A sharp maximum law). *Fix $s > 0$. Let $f(z) = az + O(z^2)$ and $g(z) = bz + O(z^2)$, where $a, b \neq 0$. Then*

$$\tau_s(f \circ g) \leq \max\{\tau_s(g), |b|\tau_s(f)\}. \quad (8.1)$$

If $\tau_s(g) \neq |b|\tau_s(f)$, equality holds. The equality assertion also applies when exactly one of the competing types is infinite.

8.1 The finite composition majorant

We first prove the inequality for tangent-to-identity inputs of finite type. Choose $A > \tau_s(f)$ and $B > \tau_s(g)$ and constants $C, D > 0$ with

$$|f_k| \leq CA^{k-1}((k-1)!)^s, \quad |g_k| \leq DB^{k-1}((k-1)!)^s \quad (k \geq 2). \quad (8.2)$$

In this section $P_0(m, \ell)$ denotes (3.2) with $\nu = 0$, and $K = K(s, 0)$. Write $g = z(1 + v)$. The terms $k = 1$ and $k = n$ in $[z^n]f(g)$ contribute g_n and f_n . In every other term, use $\ell \geq 1$ nonlinear factors of v in $(1 + v)^k$. Their total extra degree is $n - k$, and their contribution is bounded by

$$CA^{k-1}((k-1)!)^s \binom{k}{\ell} D^\ell B^{n-k} P_0(n - k, \ell). \quad (8.3)$$

Let $T = \max(A, B)$. By theorem 3.1,

$$P_0(n - k, \ell) \leq K^{\ell-1}((n - k - \ell + 1)!)^s.$$

Both factorial arguments $k - 1$ and $n - k - \ell + 1$ are positive, and sum to $n - \ell$. Hence

$$(k - 1)!(n - k - \ell + 1)! \leq (n - \ell - 1)!.$$

Also $A^{k-1}B^{n-k} \leq T^{n-1}$ and $\sum_k \binom{k}{\ell} \leq \binom{n+1}{\ell+1}$. Summing (8.3) proves

$$|[z^n]f(g)| \leq |f_n| + |g_n| + CT^{n-1}((n-1)!)^s S_n, \quad (8.4)$$

$$S_n = \sum_{\ell=1}^{n-2} D^\ell K^{\ell-1} \binom{n+1}{\ell+1} \left(\frac{(n-\ell-1)!}{(n-1)!} \right)^s. \quad (8.5)$$

This is a finite inequality valid without positivity or convergence.

For $\ell \leq n/2$, the factorial ratio is at most $(n/2)^{-s\ell}$ and the binomial factor at most $(2n)^{\ell+1}/(\ell+1)!$. For larger ℓ , the factorial ratio has at least $\lfloor n/2 \rfloor$ factors of order n^{-s} , while all other factors together are at most exponential in n . Therefore

$$S_n \leq C_1 n \exp(C_2 n^{\max(1-s, 0)}) + \exp(-cn \log n + O(n)). \quad (8.6)$$

The right side has n th root tending to 1. Thus $\tau_s(f \circ g) \leq T$. Sending $A \downarrow \tau_s(f)$ and $B \downarrow \tau_s(g)$ proves the tangent inequality. If an input type is infinite the upper bound is automatic.

For arbitrary nonzero linear coefficients, put $\hat{g} = g/b$ and $F_b(z) = f(bz)/(ab)$. These are tangent to the identity, and $f \circ g = ab(F_b \circ \hat{g})$. Since $\tau_s(F_b) = |b|\tau_s(f)$ and $\tau_s(\hat{g}) = \tau_s(g)$, (8.1) follows.

8.2 Why unequal types cannot cancel

Let $h = f \circ g$, and abbreviate

$$X = |b|\tau_s(f), \quad Y = \tau_s(g), \quad Z = \tau_s(h).$$

Apply the already proved inequality to $f = h \circ g^{(-1)}$ and $g = f^{(-1)} \circ h$, using [theorem 7.2](#). One obtains

$$X \leq \max(Y, Z), \quad Y \leq \max(X, Z), \quad Z \leq \max(X, Y). \quad (8.7)$$

If $X \neq Y$, these three inequalities force $Z = \max(X, Y)$. This also proves the extended-real assertion. It completes the proof of [theorem 8.1](#).

Example 8.2 (Both equality and strict inequality are necessary). Let $f(z) = z + \sum_{n \geq 2} A^{n-1}((n-1)!)^s z^n$ and define g similarly with $B \neq A$. Then their types are A, B , and the composition has type $\max(A, B)$. On the other hand, taking $g = f^{(-1)}$ gives two equally typed divergent inputs whose composition is the identity. Its type is zero. Thus an unconditional equality in (8.1) would be false.

Corollary 8.3 (A type filtration). *For each finite $T \geq 0$, the tangent-to-identity formal series with $\tau_s(f) \leq T$ form a group under composition. If the inner and outer types are different, the larger type survives composition.*

Proof. Use [theorems 7.2](#) and [8.1](#); the identity has type zero. □

Remark 8.4 (Analytic coordinate changes). For an analytic invertible $\phi(z) = cz + O(z^2)$,

$$\tau_s(\phi^{(-1)} \circ f \circ \phi) = |c|\tau_s(f).$$

At positive type this follows from the unequal-type equality, since an analytic series has zero s -type for $s > 0$. At zero type it follows from the upper bound. This recovers the factorial specialization of the coordinate law already present in [\[3\]](#); it is a consequence, not an additional novelty claim.

9 A positive-forward obstruction to inverse root limits

Theorem 9.1 (Full forward roots do not force full inverse roots). *For every $s > 0$ and $A > 0$, there is a tangent-to-identity formal series F with strictly positive coefficients in every positive degree and*

$$\lim_{n \rightarrow \infty} \left(\frac{[z^n]F}{((n-1)!)^s} \right)^{1/n} = A, \quad (9.1)$$

whose inverse has infinitely many zero coefficients and satisfies

$$\liminf_{n \rightarrow \infty} \left(\frac{|[z^n]F^{(-1)}|}{((n-1)!)^s} \right)^{1/n} = 0, \quad \limsup_{n \rightarrow \infty} \left(\frac{|[z^n]F^{(-1)}|}{((n-1)!)^s} \right)^{1/n} = A. \quad (9.2)$$

Proof. Define the sparse series

$$H(z) = z - \sum_{\substack{n \geq 2 \\ n \text{ even}}} A^{n-1}((n-1)!)^s z^n, \quad F = H^{(-1)}. \quad (9.3)$$

The inverse equation is

$$F = z + \sum_{\substack{m \geq 2 \\ m \text{ even}}} A^{m-1}((m-1)!)^s F^m.$$

Its recursive construction has nonnegative coefficients, and the term AF^2 alone forces strict positivity in every positive degree. In particular $F_2 = A$.

For even n , the term with $m = n$ yields

$$F_n \geq A^{n-1}((n-1)!)^s.$$

For odd $n \geq 3$, use $m = n-1$ and choose exactly one quadratic term in F^{n-1} :

$$F_n \geq A^{n-2}((n-2)!)^s(n-1)F_2 = A^{n-1}(n-1)((n-2)!)^s.$$

After normalization by $((n-1)!)^s$, the latter lower bound is $A^{n-1}(n-1)^{1-s}$. Thus both parity subsequences have normalized root liminf at least A .

The sparse series H has type A , and [theorem 7.2](#) gives $\tau_s(F) = A$. The upper limsup for F is therefore A , proving (9.1). Since $F^{(-1)} = H$, the odd coefficients vanish and the even normalized roots tend to A . This proves (9.2). $\square$

This example is deliberately not an exponential-feedback kernel. It shows exactly what a theorem based only on positivity and a forward root limit cannot establish. Eventual nonvanishing, a sign law, or a full root limit for a particular feedback inverse needs additional structure specific to that kernel.

The restriction $s > 0$ throughout the type theory is essential. At $s = 0$, $f = z - z^2$ has type zero, whereas its inverse has coefficients $\frac{1}{n} \binom{2n-2}{n-1}$ and type 4. Positive factorial growth is what makes the nonlinear overhead subexponential.

10 Reproducible calculations and what they establish

10.1 An exact coefficient algorithm

Let $f(z) = z(1 + \sum_{j \geq 1} v_j z^j)$ and, for a fixed n , write $(1 + \sum v_j z^j)^{-n} = \sum b_m z^m$. Differentiating formally gives

$$b_0 = 1, \quad b_m = -\frac{1}{m} \sum_{j=1}^m \{m + (n-1)j\} v_j b_{m-j}, \quad [z^n]f^{(-1)} = \frac{b_{n-1}}{n}. \quad (10.1)$$

For integer input, the implementation checks exact divisibility in both steps rather than silently rounding. Composition is implemented by an independent truncated polynomial multiplication routine.

The recorded standard-library run used degree 45 and performed **5,338 scalar equalities or inequalities**. It checked both inverse identities for four factorial models, all 512 sign assignments in a degree-ten coefficient box, the sparse-inverse counterexample and its lower bounds, and the identity $(f \circ g)^{\langle -1 \rangle} = g^{\langle -1 \rangle} \circ f^{\langle -1 \rangle}$ for two factorially divergent inputs. These are finite exact checks; they do not certify a limit.

For $s = C = A = 1$ and $\nu = 0$, the initial coefficients of G_* are

$$1, 1, 4, 21, 128, 862, 6264, 48549, 399352, 3484746.$$

For the sparse inverse in (9.3) at $s = A = 1$, the forward coefficients begin

$$1, 1, 2, 11, 50, 330, 1956, 15459, 109322, 1040414.$$

The complete recorded coefficients and checks are in `data/recorded/verification.json`.

10.2 Positive-term diagnostics, not interval certificates

For the extremizer with $A = 1$, normalize the recurrence by w_m . Writing $B_m = b_m/w_m$ for $m \geq 1$ gives

$$B_m = Cn + \frac{C}{m} \sum_{j=1}^{m-1} \{m + (n-1)j\} \frac{w_j w_{m-j}}{w_m} B_{m-j}, \quad R_n = \frac{B_{n-1}}{nC}. \quad (10.2)$$

This avoids building the final huge factorial before normalization. All summands are positive. The diagnostic implementation uses extended-range `longdouble`, rejects environments where that format has only ordinary double range, and checks finiteness. Three cases at degree 80 were independently recomputed at 80 decimal digits with `mpmath`; relative differences were below $5 \cdot 10^{-18}$. Neither computation uses directed rounding.

s	n	$\log R_n$	$\log R_n/n^{1-s}$	$\log R_n - n^{1-s}$
0.25	320	193.909447	2.562928	118.250119
0.25	1280	363.057932	1.696557	149.061035
0.5	320	20.613024	1.152303	2.724481
0.5	1280	38.185907	1.067329	2.408819
0.75	320	4.404885	1.041471	0.175399
0.75	1280	6.062521	1.013563	0.081126
1	320	1.015916	1.015916	0.015916
1	1280	1.003924	1.003924	0.003924
1.25	320	0.238729	1.009699	0.002293
1.25	1280	0.167556	1.002219	0.000371

Table 2: Recorded diagnostics for $C = 1$, $\nu = 0$. Every displayed entry is a rounded floating-point diagnostic. At $s = 1/2$, the last column tends to $3/4 + \sqrt{2} = 2.16421356\dots$; convergence is visibly not complete at the displayed degrees.

At $s = 1/4$, the ratio in the fourth column is still about 1.70 at $n = 1280$, although its proved limit is 1. This is an important caution: a leading logarithmic theorem need not provide a useful small-degree approximation. We do not use the table to assert an onset or error rate.

t	$s = 1 + t/\log 1280$	observed R_{1280}	$\exp(e^{-t})$
-2	0.720460	1823.607153	1618.177992
-1	0.860230	15.460266	15.154262
0	1.000000	2.728969	2.718282
1	1.139770	1.446060	1.444668
2	1.279540	1.145256	1.144921

Table 3: The order-one crossover with $C = 1$, $\nu = 0$. The rightmost column is the limit in (2.10), not a finite-degree enclosure.

10.3 Rebuilding without modifying recorded results

The package has no network dependency. From its root directory:

```
python code/verify.py -order 45
python code/verify.py -order 45 -diagnostics -mp-check
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact tests need only Python’s standard library. Diagnostics require NumPy, and the high-precision cross-check requires mpmath. The default output directory is `data/rerun/`, so an ordinary rerun does not overwrite the recorded evidence. The numerical tables used by this article are supplied as small TeX files generated from the recorded CSVs.

11 Implications for the transseries program

11.1 What the result adds below type preservation

A type identifies a coefficient series only up to factors whose n th roots tend to 1. At fixed positive Gevrey order, that equivalence is too coarse to control a critical-radius coefficient norm. The extremal cost quantifies the missing information. In the shifted normalization, quadratic coefficients generate approximately Cn^{1-s} small Lagrange blocks, and their accumulation produces $\exp(Cn^{1-s})$. At and above $s = 1$ their number stays bounded or tends to zero.

This has a practical implication for certified symbolic manipulation: retaining an optimal exponential constant does not by itself justify retaining a fixed coefficient prefactor. For $s < 1$, a program that returns the same A and a constant-size majorant after reversion will fail on the explicit extremizer. It must either track a subexponential envelope, allow an arbitrarily small increase of A , or exploit extra structure of its particular input.

11.2 What transfers to countable exponential feedback

The source weighted-type theorem is motivated by

$$U(q) = \sum_{j \geq 1} c_j q^j \exp(\lambda_j U(q)).$$

Once a separately established theorem places such a forward series in a particular factorial coefficient ball, the exact finite majorant (3.3) bounds its inverse. The bound is universal and requires no information about the inverse signs. The composition theorem also controls the type when two divergent formal correction maps are nested.

These statements do *not* say that a particular feedback inverse attains the extremal amplification. The extremizer has negative nonlinear input coefficients; positive-feedback inputs can exhibit substantial cancellation after inversion. Consequently, this article makes no new signed equivalent, least-term error law, or summability claim for the feedback family. Its contribution is the universal fixed-ball estimate and the sharp filtered composition law.

11.3 Formal and analytic layers remain separate

Everything in the proofs before the numerical section concerns ordinary formal coefficient degrees. An exponential transseries with accumulating actions may not have a finite degree decomposition of the kind used in (3.2). Applying these arguments to such supports requires an additional finite-coefficient or summability theorem. Similarly, a formal inverse need not be the asymptotic expansion of a chosen analytic inverse without an analytic realization theorem. Neither issue is removed by the positivity of the extremal majorant.

12 Further research questions and concrete next targets

12.1 Complete multiplicative cost below one half

For $0 < s < 1/2$, determine a finite explicit expansion of $\log R_n$ whose error is $o(1)$. The Poisson representation suggests a hierarchy of scales n^{1-rs} . Small parts of size r become visible at $s = 1/r$, but product curvature and tilted fluctuations also contribute. The formula for $s = 1/2$ proves that simply summing the small-part intensities is insufficient. A useful next theorem would give the complete equivalent for $1/3 < s < 1/2$, with its constant at $s = 1/3$. No coefficient for that further expansion is asserted here.

12.2 Uniform transitions at the lower thresholds

Construct analogues of (2.10) around $s = 1/r$ for $r \geq 2$. The variables $(s - 1/r) \log n$ are natural candidates, but one must first subtract all larger contributions. A successful theorem should specify both a uniform remainder and the normalization under which a finite limiting profile appears.

12.3 Replacing factorials by a local-quotient theory

For a log-convex weight M , seek hypotheses under which the critical inverse cost is controlled sharply by nM_{n-1}/M_n , after the appropriate index convention is fixed. In the present model the ratio is asymptotic to n^{-s} , explaining the scale n^{1-s} . Slowly varying perturbations, abrupt quotient changes, and factorial weights with logarithmic deficits need separate uniformity analysis. The repository's general zero-loss theorem alone is not enough to identify their extremal subexponential constants.

12.4 Two divergent inputs for general weights

Extend theorem 8.1 beyond factorial powers. A candidate hypothesis is a weighted two-part convolution estimate with subexponential constants uniform in the number of parts. Determine whether log-convexity and superexponential root growth alone suffice, or construct a counterexample. Equal-type cancellation must remain part of the statement.

12.5 Typical inputs versus the extremizer

Classify coefficient sign patterns or asymptotic regularity assumptions under which $|[z^n]f^{(-1)}|$ is asymptotic to a fixed fraction of the extremal coefficient. An opposite target is a sharp suppression theorem for positive forward coefficients. The sparse example shows that a root-limit hypothesis by itself cannot provide such a classification.

12.6 Exact remainder bounds at finite degree

Turn the localization proof into a computable error certificate for (2.8) and (2.9), with an explicit optimal cutoff h and a rigorous bound for every remainder. The finite sum (3.3) already gives an exact extremum, but the present asymptotic proof does not supply an optimized, practical onset bound. Directed-rounding evaluation of the positive recurrence would provide a complementary finite-degree certificate.

12.7 Several variables and operator-valued coefficients

Define a coefficient ball for homogeneous polynomial maps or bounded multilinear operators with an invertible linear part. Determine which normed extremal problem replaces the one-variable negative series and whether the threshold $s = 1$ persists without dimension-dependent exponential loss. A type-level theorem and a sharp fixed-ball theorem should be treated as different objectives.

12.8 A realistic formalization sequence

A Lean formalization can begin with the finite identities (3.1), (3.3), and (8.4). The next independent component is the factorial concentration and convolution library, including a summable uniform bound for K . A third component proves the tail estimates and the type maximum law. Only after those finite and limsup statements are established is it reasonable to formalize localization, the Poisson normalization, and uniform asymptotics. The present Python checks should remain explicitly separate from that program.

13 Conclusion

The Gevrey version of the repository's optimal-overhead question has a sharp answer at the leading subexponential scale: a single extremizer attains $\log R_n \sim Cn^{1-s}$ for $0 < s < 1$. Its full multiplicative behavior is determined here at and above one half, including the boundary constant, while the uniform transition at order one separates fixed-radius boundedness from type preservation. Two divergent formal maps obey a maximum law at the level of Gevrey type, and unequal types cannot cancel. Full root limits require additional information, as the explicit positive-forward counterexample demonstrates.

These are extensions of existing repository results, not replacements for the already established zero-loss theorem. They isolate what must be tracked beyond an optimal exponential constant and supply finite formulas suitable for both implementation and subsequent formalization.

A A compact proof audit

The logical dependencies can be checked without reading the numerical section. Formal Lagrange inversion implies the exact extremizer formula. The two-part convolution lemma implies the

length tail bound. Capped factorial concentration plus that length tail implies localization. Localization and an elementary negative-binomial upper bound, paired with the all-size-one lower bound, prove the sharp logarithmic cost. Localization at a shrinking cutoff and a finite Poisson normalization prove the multiplicative equivalent. The order-one-half correction uses one extra product expansion and a separately proved exponential integrability estimate. No empirical observation is used in these steps.

The type-invariance proposition uses only the coarse convolution bound and inversion in both directions. The composition theorem uses the finite composition majorant for its upper bound and type invariance for its noncancellation conclusion. The root-limit counterexample uses positivity, two explicit coefficient lower bounds, and type invariance.

The potentially delicate points are therefore exposed: uniformity in the convolution constant; entropy control with $t \log(en/t)$ rather than the too-coarse $t \log n$ in the fixed-fraction localization proof; uniqueness of the large part; the distinction between an absolute $o(1)$ discarded contribution and a relative error; and the exponential-integrability argument at $s = 1/2$. These points are all proved explicitly in the main text.

B Package contents and provenance

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Source and compiled article.
<code>README.md</code>	Scope, main results, and rebuild commands.
<code>code/verify.py</code>	Exact finite checks and optional diagnostics.
<code>code/render_tables.py</code>	Regenerate the printed tables from CSV files.
<code>data/recorded/verification.json</code>	Recorded exact-check ledger and high-precision comparisons.
<code>data/recorded/diagnostics.csv</code>	Positive-term extremal computations.
<code>data/recorded/crossover.csv</code>	Order-one transition diagnostics.
<code>data/*_table.tex</code>	Tables generated from the recorded CSVs.
<code>data/build_validation.json</code>	Build and layout validation record.
<code>provenance.json</code>	Repository snapshot, source paths, and dependency boundaries.

No repository source file is changed by this package. No claim is made that the full ProveIt corpus was exhaustively searched. In particular, concurrently arriving or unintegrated archives are not certified to be disjoint from this article. The explicit crosswalk in [section 1](#) records the actual target and avoids claiming the already settled inverse-type question as a new result.

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